

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a33d9f05cf6d7be75.jsonl`  
- SHA-256: `12c04f3aa380de27e60a4e6c82e714451b93231183da4abf36f5255af84224f6`  
- Source modified: `2026-05-31T04:26:18+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:25:50.210Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"OmniDocBench is a CVPR 2025 document-parsing benchmark by OpenDataLab (Shanghai AI Lab) covering 1,651 PDF pages across 10 document types ? and it explicitly includes financial reports as one of the document categories, making its table-recognition results directly relevant to bank-statement parsing."

****Source:**** <https://github.com/opendatalab/OmniDocBench> (primary)

****Supporting quote:**** "It encompasses a wide range of content, including academic papers, financial reports, newspapers, textbooks, and handwritten notes."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:25:53.277Z)

I'll verify this claim about OmniDocBench. Let me start by examining the primary source and searching for contradicting evidence.

3. assistant (2026-05-31T04:25:56.586Z)

Let me fetch the primary source and search for the CVPR 2025 claim simultaneously.

4. user (2026-05-31T04:26:03.034Z)

OmniDocBench Details

**** (1) CVPR 2025 Paper/Benchmark: ****

Yes. The page explicitly states: "[CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation" in the title and confirms "OmniDocBench has been accepted by CVPR 2025!"

**** (2) Creator: ****

OpenDataLab created this benchmark (affiliated with Shanghai AI Lab, based on context).

**** (3) PDF Pages: ****

"This benchmark includes 1651 PDF pages" according to the benchmark introduction section.

**** (4) Document Types: ****

The benchmark covers "10 document types" as stated multiple times in the overview.

**** (5) Document Categories - Exact List: ****

The page specifies: "academic papers, financial reports, newspapers, textbooks, and handwritten notes." The term "financial reports" appears in the attribute classification section as "research_report # Research reports and financial reports," confirming financial reports are explicitly included as a category.

5. user (2026-05-31T04:26:04.673Z)

Web search results for query: "OmniDocBench CVPR 2025 OpenDataLab document parsing benchmark 1651 pages"

Links: [{"title": "Activity · opendatalab/OmniDocBench", "url": "https://github.com/opendatalab/OmniDocBench/activity"}, {"title": "Releases · opendatalab/OmniDocBench", "url": "https://github.com/opendatalab/OmniDocBench/releases"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub", "url": "https://github.com/opendatalab/OmniDocBench"}, {"title": "translate.google.com/opendatalab/OmniDocBench?_x_tr_sl=en&_x_tr_tl=es&_x_tr_hl=es&_x_tr_pto=tc", "url": "https://translate.google.com/opendatalab/OmniDocBench?_x_tr_sl=en&_x_tr_tl=es&_x_tr_hl=es&_x_tr_pto=tc"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub", "url": "https://github.com/opendatalab/OmniDocBench"}, {"title": "GitHub - marklabz/omnidocbench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub", "url": "https://github.com/marklabz/omnidocbench"}, {"title": "OmniDocBench/README_zh-CN.md at main · opendatalab/OmniDocBench", "url": "https://github.com/opendatalab/OmniDocBench/blob/main/README_zh-CN.md"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/opendatalab/OmniDocBench?tab=readme-ov-file"}, {"title": "OmniDocBench is Saturated, What's Next for OCR Benchmarks?", "url": "https://www.llamaindex.ai/blog/omnidocbench-is-saturated-what-s-next-for-ocr-benchmarks"}]

Based on the search results, here's what I found about OmniDocBench:

OmniDocBench has been accepted by CVPR 2025 and is a comprehensive benchmark for document parsing that has developed an evaluation methodology based on document component segmentation and matching, providing metric calculations for four major modules: text, tables, formulas, and reading order.

OmniDocBench (CVPR 2025) is a benchmark published by Ouyang et al. at OpenDataLab, and its document dataset spans nine different document sources, ranging from academic papers to textbooks to handwritten notes.

The benchmark has evolved over time, with the original benchmark including 981 PDF pages, covering 9 document types, 4 layout types, and 3 language types, and 296 new pages were added, with samples chosen to cover more challenging scenario categories including complex nested tables, dense mathematical formula layouts, and unconventional layout structures.

However, the search results do not contain specific information about the "1651 pages" figure you mentioned in your query.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:26:09.434Z)

Interesting. The WebFetch confirms most of the claim, but the search result reveals an important nuance: the page count appears to have evolved. The search says the "original benchmark including 981 PDF pages, covering 9 document types" with 296 new pages added later. The claim says "1,651 PDF pages across 10 document types." Let me dig into the exact current numbers and verify the "financial reports as a category" point, which is the load-bearing part of the claim for bank-statement relevance.

Let me check the README more precisely and look at the document category structure, plus check whether "financial reports" is a distinct category or lumped with "research reports."

7. user (2026-05-31T04:26:16.796Z)

Web search results for query: "OmniDocBench document types list "financial report" OR "research report" nine categories taxonomy"

Links: [{"title": "OmniDocBench: A Real-World, Fine-Grained Benchmark for Fair and Comprehensive PDF Document Parsing Evaluation | PaperCodex", "url": "https://www.papercodex.com/omnidocbench-a-real-world-fine-grained-benchmark-for-fair-and-comprehensive-pdf-document-parsing-evaluation/"}], [{"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/opendatalab/OmniDocBench"}, {"title": "omnidocbench · Datasets at Hugging Face", "url": "https://huggingface.co/datasets/opendatalab/OmniDocBench"}, {"title": "samiuc/omnidocbench · Datasets at Hugging Face", "url": "https://huggingface.co/datasets/samiuc/omnidocbench"}, {"title": "[Literature Review] OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations", "url": "https://www.themoonlight.io/en/review/omnidocbench-benchmarking-diverse-pdf-document-parsing-with-comprehensive-annotations"}, {"title": "GitHub - MASLINQUE/omnidocbench: A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/MASLINQUE/omnidocbench"}, {"title": "OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations", "url": "https://arxiv.org/html/2412.07626v1"}, {"title": "GitHub - marklabz/omnidocbench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/marklabz/omnidocbench"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation", "url": "https://github.com/opendatalab/OmniDocBench?tab=readme-ov-file"}, {"title": "OmniDocBench Leaderboard: PDF Document Parsing SOTA | CodeSOTA | CodeSOTA", "url": "https://www.codesota.com/browse/computer-vision/document-parsing/omnidocbench"}]

Based on the search results, I found information about OmniDocBench's document type taxonomy. OmniDocBench includes 9 document types: academic papers, financial reports, newspapers, textbooks, handwritten notes, exam papers, magazines, research reports, and PPT-to-PDF conversions.

The search results show that both "financial report" and "research report" are included as separate categories in this nine-category taxonomy for document types within the OmniDocBench benchmark, which is designed for evaluating diverse PDF document parsing capabilities.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-05-31T04:26:17.591Z)

OmniDocBench Dataset Statistics

(1) Total PDF Pages:

"This benchmark includes 1651 PDF pages" ? this is the exact current count stated in the benchmark introduction.

(2) Document Types:

The benchmark covers **10 document types**, as explicitly stated: "covering 10 document types, 5

layout types, and 5 language types."

****(3) Full Document Type Categories:****

The verbatim taxonomy lists these categories under the `data\_source` attribute:

- academic_literature
- PPT2PDF
- book
- colorful_textbook
- exam_paper
- note
- magazine
- research_report
- newspaper
- (The list shows 9 items, though the benchmark claims 10 types)

****(4) Financial Reports Classification:****

Financial reports are **combined** with research documents under a single category label:

"research_report" is defined as "Research reports and financial reports." They are not separate distinct categories in the taxonomy.

9. assistant (2026-05-31T04:26:17.997Z)

You've hit your session limit · resets 1:30pm (Asia/Calcutta)